

Mid Sem Syllabus

Deep Learning (DL)

Paper: Deep Learning

Feed Forward Neural Network

Introduction to Neural Networks

- The Perceptron
- Linear Models
- The XOR Problem
- Shallow Neural Network
- Forward Propagation Network

Activation Functions

- Sigmoid
- Tanh
- ReLU and Leaky ReLU
- ELU
- Softmax Function
- Universal Approximation Theorem
- Depth Trade-off
- Width Trade-off

Loss Functions

- Maximum Likelihood Estimation
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Cross-entropy Loss

Backpropagation

- Computational Graphs
- Chain Rule
- Forward Pass
- Backward Pass
- Backpropagation Algorithm
- Gradient Flow
- Vanishing Gradients
- Exploding Gradients

Neural Network Optimization

Types of Gradient Descent

- Vanilla Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent

Learning Rate Optimization

- Fixed Learning Rate
- Learning Rate Scheduling

Momentum-Based Methods

- Exponentially Weighted Moving Average (EWMA)
- SGD with Momentum
- Nesterov Accelerated Gradient (NAG)

Adaptive Learning Rate Methods

- AdaGrad
- RMSProp
- Adam

Regularization Techniques in Neural Networks

Bias-Variance Analysis

- Underfitting
- Overfitting
- Bias-Variance Tradeoff
- Double Descent

Regularization Methods

- L1 Regularization
- L2 Regularization
- Dropout
- Batch Normalization
- Early Stopping

Normalization & Initialization

- Zero Initialization
- Constant Initialization
- Random Initialization
- He Initialization
- Xavier Initialization

Convolutional Neural Networks

CNN Fundamentals

- Filters / Kernels
- Stride
- Padding
- Feature Maps
- Pooling
- Receptive Field
- Translation Invariance

Classic Architectures

- Lenet
- VGG, Alexnet, Inception net

Deep CNN Architectures

- The Shattered Gradients Problem
- Residual Connections
- ResNet

Mode of Exam :- Online

Calculators are allowed